

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 13



Non Regular Languages

We will establish a property that all *regular languages* must possess.

This property is called the *pumping lemma*.

If a language does not have this property, then it must be *nonregular*.

Pumping Lemma for *Regular* Languages

Let L be a *regular* language.

Then there exists an integer $n \geq 1$, called the *pumping length*, such that the following holds:

For *every string* $w \in L$, with $|w| \geq n$, *can be rewritten/decomposed* as $w = xyz$, such that

1. $y \neq \epsilon$

2. $|xy| \leq n$, and

3. $xy^iz \in L$ *for each* $i \geq 0$

By replacing the portion y in w by **zero or more copies** of it, the resulting string is still in the language L .

First non regular language

First non regular language

$L = \{0^n 1^n : n \geq 0\}$ is not a regular language

Suppose L was regular

Let $n \geq 1$ be the pumping length as in Pumping Lemma

Consider the string $w = 0^n 1^n$

Key is to choose the right string.

Then $w \in L$ and $|w| = 2n \geq n$

For example $w = 0^{n/2} 1^{n/2}$ will not help much

Hence, by the pumping lemma, w can be rewritten/decomposed as

$w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Consider any decomposition of w as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Since $|xy| \leq n$, the string y contains only 0's

Moreover, since $y \neq \epsilon$, y contains at least one 0 or $y = 0^r$ for some $r > 0$

But then for $i = 0$, $xy^i z = xz = 0^{n-r} 1^n \notin L$, a contradiction.

Non regular language

$L = \{w \in \{0, 1\}^* : \text{the number of 0's in } w \text{ equals the number of 1's in } w\}$

Observe

Exactly the same arguments works as in proving

$L' = \{0^n 1^n : n \geq 0\}$ is not a regular language

OR

Closure Properties can be used to prove language is not regular.

Suppose L was regular.

Note $L \cap 0^*1^* = L'$

And by *closure property* under *intersection* of regular languages, L' would a regular language.

A contradiction.

Non regular language

Let $L = \{ww : w \in \{0, 1\}^*\}$

Suppose L was regular

Let $n \geq 1$ be the pumping length as in Pumping Lemma

Consider the string $w = 0^n 1 0^n 1$

Then $w \in L$ and $|w| = 2n + 2 \geq n$

Hence, by the pumping lemma, w can be rewritten/decomposed as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Consider any decomposition of w as $w = xyz$, where $y \neq \epsilon$, $|xy| \leq n$, and $xy^i z \in L$ for each $i \geq 0$

Then $x = 0^q$ and $y = 0^r$ where $q \geq 0$ & $r > 0$

Let $i = 2$, then $xy^2 z = xyyz$ has more 0's before the first 1 than after 1.

Therefore $xy^2 z \notin L$. A contradiction.

FSA State Minimization

$$L = (ab \cup ba)^*$$

And the FSA accepting L is M:

Consider the states q_7 & q_8

Observe:

These states are *unreachable*

Because there is no path from the start state to it.

Simplest State Optimization for an DFA:

Remove all *unreachable states* & all transitions in and out of them.

Next Observe

For the states q_4 & q_6 precisely the **same strings** lead M to *acceptance*.

We may think these as '*equivalent*' and **merge** them.

